

IV Year B.Tech CSE I Semester**L T P C****Course Code: 127EY****3 0 0 3****DATA SCIENCE USING R**

(Professional Elective-4)

Prerequisites: Database Management Systems.**Course Objectives:**

1. To provide the knowledge and expertise to become a proficient data scientist.
2. Demonstrate an understanding of statistics and machine learning concepts that are vital for data science.
3. To understand statistically to analyze a dataset.
4. Critically evaluate data visualization based on regression and time series in R.
5. To learn association rule mining and text mining algorithms for analyzing the dataset.

UNIT 1: (~9 Lecture Hours)**Introduction to Data Science:** Basics of data Science, Role in a Data Science Project, stages of a data science project-defining the goal, data Collection and management.**Introduction to R:** Introduction, Downloading and Installing R, IDE and Text Editors, Handling, Packages in R, Working with Directory, Data Types, Commands for Data Exploration.**Loading and Handling Data in R:** Introduction, working with data from files, Working with relational databases.**UNIT 2: (~9 Lecture Hours)****Exploring Data in R:** Introduction, Data Frames, R Functions for Understanding Data in Data Frames, Load Data Frames, Exploring Data, Data Summary, Finding the Missing Values, Invalid Values and Outliers, Descriptive Statistics, Spotting Problems In Data with Visualization.**Managing Data in R:** Cleaning data, sampling for Modeling and Validation.**UNIT 3: (~9 Lecture Hours)****Modeling Methods:** Choosing and Evaluating models: Mapping problems to machine learning tasks, Evaluating Models, Validating Models.**Memorization Methods:** KDD and KDD Cup 2009, Building Single variable models, Building models using many variables.**UNIT 4: (~9 Lecture Hours)****Linear Regression Using R:** Introduction, Understanding Linear Regression, Building a Linear Regression Model, Making predictions.

Logistic Regression: Introduction, What Is Regression?, Introduction to Generalized Linear Model, Logistic Regression, Introduction, Understanding Logistic Regression, Building a Logistic Regression Model , Making predictions.

Unsupervised Methods: Cluster analysis – Distances, Preparing the data, Hierarchical clustering, The k-means algorithm.

UNIT 5: (~9 Lecture Hours)

Time Series in R: Introduction, What Is Time Series Data, Similarity Search in Time series Data, Regression and trend Analysis in Time series data, Reading Time Series Data, Plotting Time series Decomposing Time Series Data-Decomposing Seasonal Data, decomposing Non Seasonal data, Seasonal adjusting, Forecasts using Exponential smoothing, ARIMA models.

Text Books:

1. Nina Zumel and John Mount, Practical Data Science with R, 2nd Edition, Manning publications.
2. Seema Acharya, Data Analytics using R, 1st Edition, McGraw Hill Education.

Reference Books:

1. Jure Leskovek, Anand Rajaraman and Jeffrey Ullman, Mining of Massive Datasets, v2.1, Cambridge University Press.
2. Foster Provost and Tom Fawcett, Data Science for Business, What You Need to Know about Data Mining and Data-Analytic Thinking, O'Reilly.
3. Cathy O Neil and Rachel Schutt, Doing Data Science, Straight Talk from the Frontline, O'Reilly.
4. Samir Madhavan, Mastering Python for Data Science, Packt Publishing, 2015.

Online Resources:

1. <http://datasciencemasters.org/>.
2. <http://learnds.com/>
3. <https://www.datascienceweekly.org/>

Course Outcomes:

After completion of the course, students will be able to

1. Understand and apply basic concepts of R for data science.
2. Learn and apply the fundamental concepts of data science.
3. Design suitable linear regression models to extract useful information.
4. Develop and analyze regression models for the given data.
5. Handle time series data using ARIMA models.
6. Analyze various association rule mining and text mining algorithms for better understanding.

IV Year B.Tech CSE I Semester**L T P C****Course Code: 12753****0 0 2 1****DATA SCIENCE USING R LAB**

(Professional Elective-4)

Prerequisites: Database Management Systems**Course Objectives:**

1. Understand the R Programming Language.
2. Exposure on Solving of Data Science Problems.
3. Understand the classification Model.
4. Understand the Regression Model.
5. Able to Understand Statistics usage in data Science.

Week 1-2:**Calculator Application:**

- a) Using with and without R objects on console.
- b) Using mathematical functions on console.
- c) Write an R script, to create R objects for calculator application and save in a specified location in disk.
- d) R Packages: tidy, ggplot2, ggraph, glue. shiny.

Week 3:**Descriptive Statistics:**

- a) Write an R script to find basic descriptive statistics using summary.
- b) Write an R script to find subset of dataset by using subset.

Week 4:**Reading and Writing Different Types of Datasets:**

- a) Reading different types of data sets (.txt, .csv) from web and disk and writing in file in specific disk location.
- b) Reading Excel data sheet in R.
- c) Reading XML dataset in R.

Week 5-6:**Visualizations:**

- a) Find the data distributions using box and scatter plot.
- b) Find the outliers using plot.
- c) Plot the histogram, bar chart and pie chart on sample data

Week 7:**Correlation and Covariance:**

- a) Develop a program to find the correlation matrix on iris data.
- b) Plot the correlation plot on dataset and visualize giving an overview of relationships among data on iris data
- c) Analysis of covariance: variance (ANOVA), if data have categorical variables on iris data.

Week 8:**Regression Model:**

- a) Import a data from web storage. Name the dataset and now do Logistic Regression to find out relation between variables that are affecting the admission of a student in an institute based on his or her GRE score, GPA obtained and rank of the student. Also check the model is fit or not. Require (foreign), require (MASS).
- b) Apply regression Model to predict the data on any dataset.

Week 9:**Multiple Regression Model:**

- a) Develop Multiple Regression model for any dataset.
- b) Apply Multiple Regression Model to predict the data on any dataset.

Week 10:**Classification Model:**

- a) Install relevant package for classification.
- b) Implement and evaluate the performance of various classifiers with different datasets.

Week 11:**Clustering Model:**

- a) Clustering algorithms for unsupervised classification.
- b) Plot the cluster data using R visualizations.

Week 12:**Time Series Model:**

- a) Plot the Time series data using R visualizations.
- b) Time series Forecasting using Exponential smoothing, ARIMA models.

Text Books:

1. Yanchang Zhao, R and Data Mining: Examples and Case Studies, Elsevier, 1st Edition, 2012.
2. Nina Zumel and John Mount, Practical Data Science with R, 2nd Edition, Manning Publications.